

Total No. of Questions :10]

SEAT No. :

P3066

[5059]-538

[Total No. of Pages : 3

B.E. (Mechanical)

GAS TURBINE & PROPULSION

(2012 Course) (End-Semester) (Semester-I) (Elective-II) (402045-A)

Time : 2½ Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Write a note on ideal and actual Brayton cycle. **[3]**

- b) In a gas turbine plant, the suction occurs at 1 bar and 27°C. The air is compressed to 4 bar. Maximum temperature is 572°C. Find the air-fuel ratio. Assume isentropic efficiency of compressor & turbine, as 85%. Take C_p for air & gases as 1.005 and 0.97 kJ/kgK, respectively, C.V = 42 MJ/kg. **[7]**

OR

Q2) Write notes on: **[10]**

- a) Rocket propulsion.
- b) Polytropic efficiency of compressor & turbine.

Q3) a) Write a note on the various parameters affecting the flight performance. **[4]**

- b) Explain the working of turbofan engine. **[6]**

OR

Q4) a) Discuss Nozzle & diffuser losses. **[5]**

- b) What do you understand by the term air-breathing engines? How are they classified? **[5]**

P.T.O.

- Q5) a)** Explain the effect of nozzle angle on work output and effect of axial discharge in an axial flow turbine. [6]
- b) In a single stage axial flow turbine (Impulse type), the expansion in the nozzle occurs from 4 bar, 650°C to a pressure of 1 bar. The discharge angle is 15°. The blade speed is 360 m/s and the gas leaves the rotor blade in axial direction with a speed of 300 m/s. Assuming a nozzle efficiency of 96%. Find: [10]
- i) Rotor blade angles at inlet and outlet.
 - ii) Utilization factor.
 - iii) Power developed for 25 kg/sec of gas flow.
 - iv) Stage efficiency.

OR

- Q6) a)** Discuss the performance curves of gas turbines. [6]
- b) Gas enters in an impulse turbine (1-stage) at a stagnation pressure of 4 bar while the exit static pressure is 1.18 bar. Stage exit velocity is 290 m/s (axial) & nozzle efficiency is 96%. The blade velocity is 320 m/s. The static head efficiency is 88%. The exit pressure from the nozzle is 1.25 bar. Find: [10]
- i) W. D. per kg of air.
 - ii) Nozzle angle.
 - iii) Blade inlet and outlet angle.
 - iv) Energy utilization factor for the rotor.
- Q7) a)** Explain the processes involved in an axial flow compressor with the help of T-s diagram. [6]
- b) Write notes on: [10]
- i) Flow coefficient & stage efficiency.
 - ii) Degree of reaction and workdone factor.

OR

Q8) Write notes on:

[16]

- a) Radial & axial flow compressors.
- b) Factors affecting pressure ratio.
- c) Inlet & Outlet velocity triangle.

Q9) a) Explain the different types of combustion chambers used in gas turbines. **[9]**

b) Write notes on: **[9]**

- i) Desirable features of combustion chamber of a gas turbine.
- ii) Combustion chamber arrangements.

OR

Q10) Write notes on:

[18]

- a) Off-design characteristics of gas turbines.
- b) Stability limits and combustion intensity.
- c) Component characteristics of compressor & turbine.

●●●●●